

Martian Surface Segmentation Using U-Net in PyTorch

By: Ary Naim UNM (anaim@unm.edu)

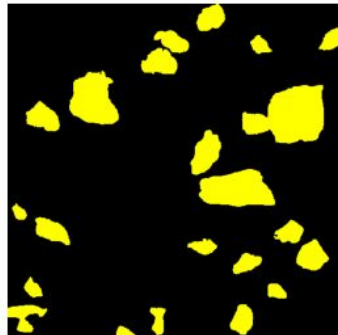
Advisor: Dr Marios Pattichis

Department of Electrical and Computer Engineering

Original Image



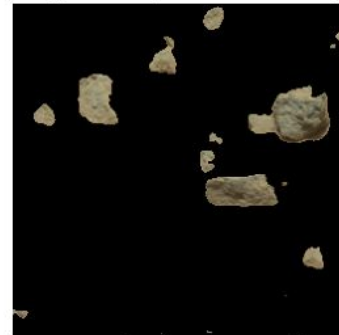
Ground Truth Mask



Predicted Mask



Segmented (Predicted Mask)



Problem

Mars remains a frontier for exploration and a stepping stone for space exploration. Due to the distance from Earth; Martian exploration may initially depend on autonomous operations that may face many hazards such as rocks and craters. I propose to implement the U-Net-only image segmentation Martian landscape segmentation for rock detection. Rocks remain a major hazard to any autonomous planetary rover.

Refer to Github: <https://github.com/naimaryan1/LunarMUnet>

Collab Notebook:

https://colab.research.google.com/drive/1aiZMAFvt1e5_HM59oQ0EtJy6HIfzMpB?usp=sharing

A Martian Dataset

- Planetary surface datasets are limited
- There is currently the limited MarsData-V2 dataset (<https://github.com/CVIR-Lab/MarsData>)
- However they claimed 300 unique images but in reality its ~30 images (30 times 10 augmentations):

Training Data

331_0331MR0013390000301024E01_DXXX_rccw.png



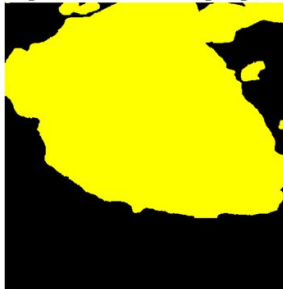
332_0332ML0013440000107964E01_DXXX_raw.png



332_0332MR0013440010301049E01_DXXX_vf.png

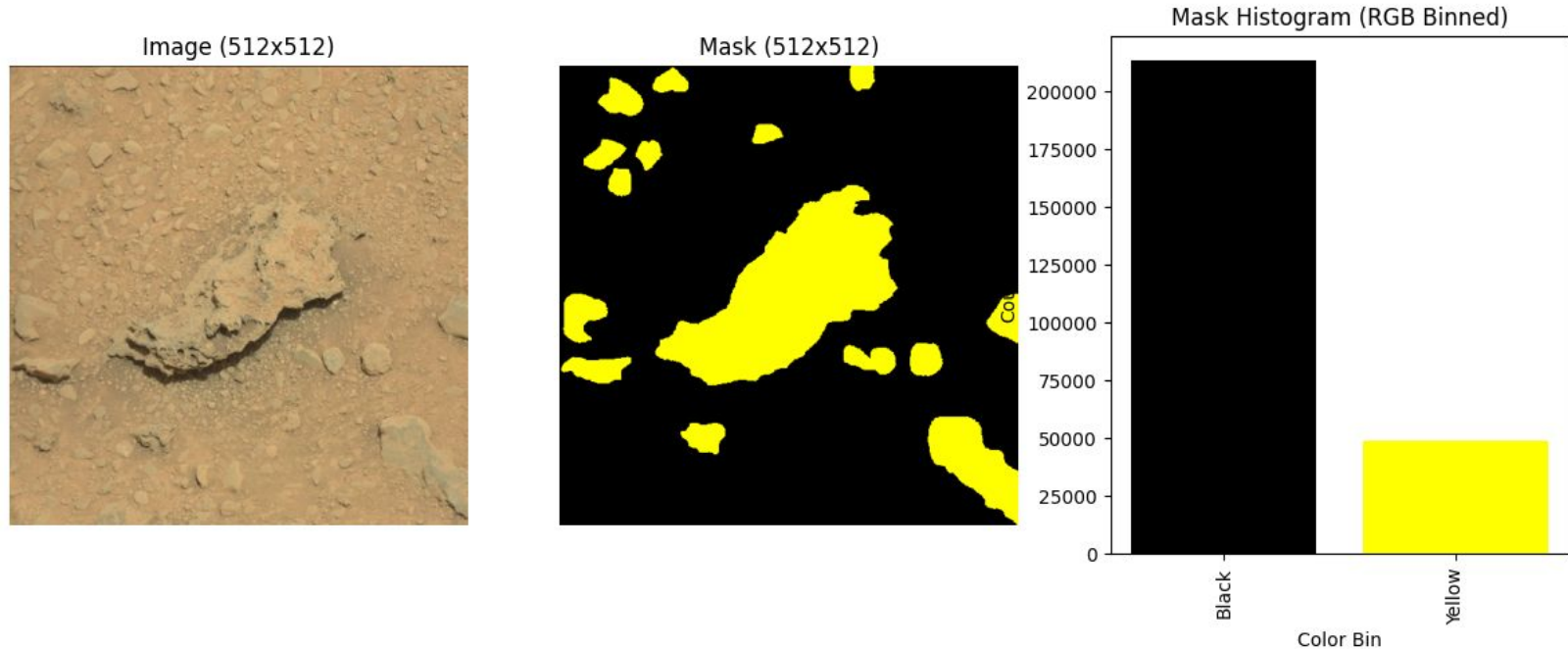


332_0332MR0013440010301049E01_DXXX_tup.png



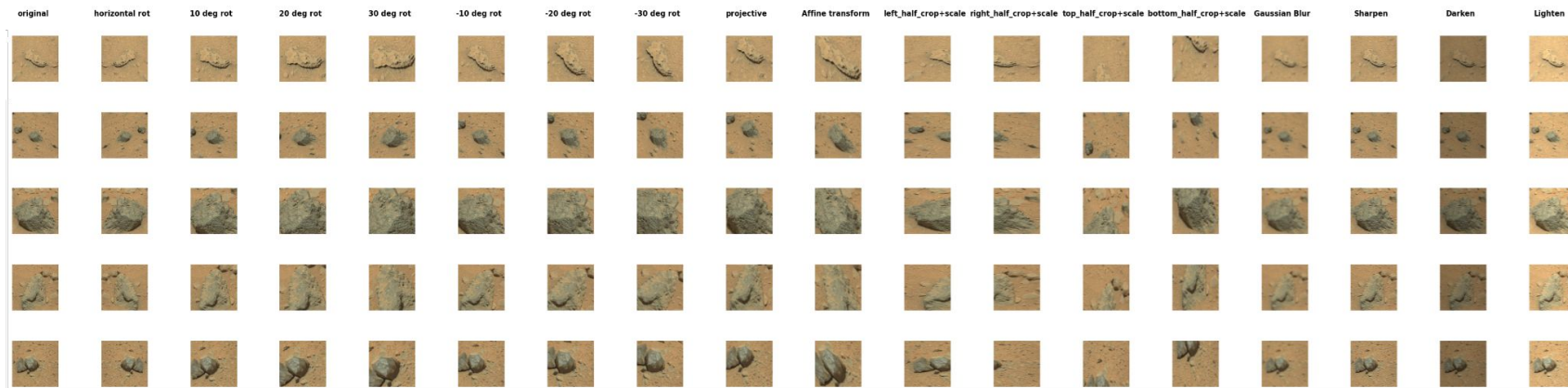
Martian - Visualize Image, Mask, and Mask's Histogram

There are 2 colors in the mask corresponding to 2 of the expected classes

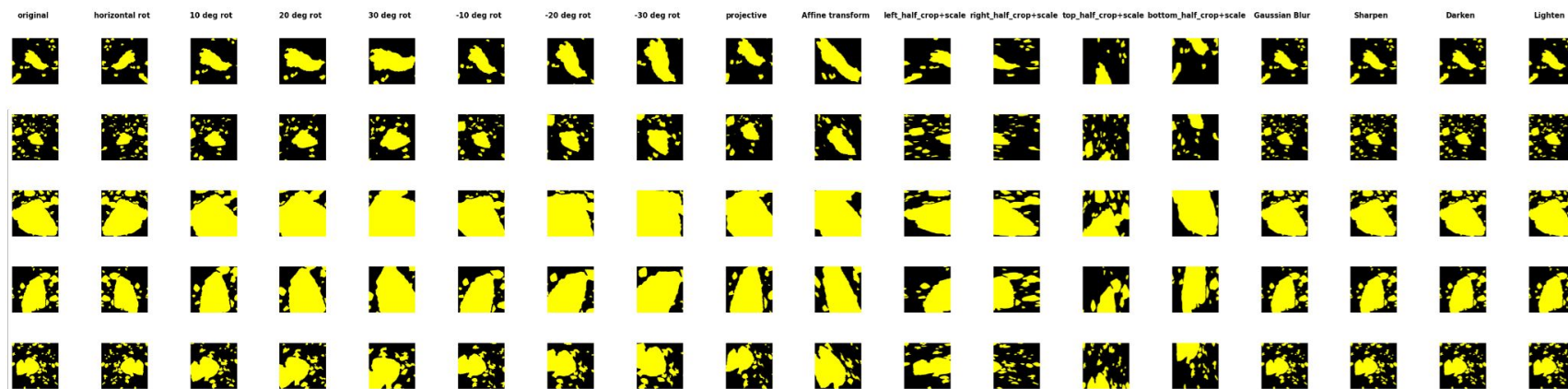


Martian Data Augmentation

Augmentations



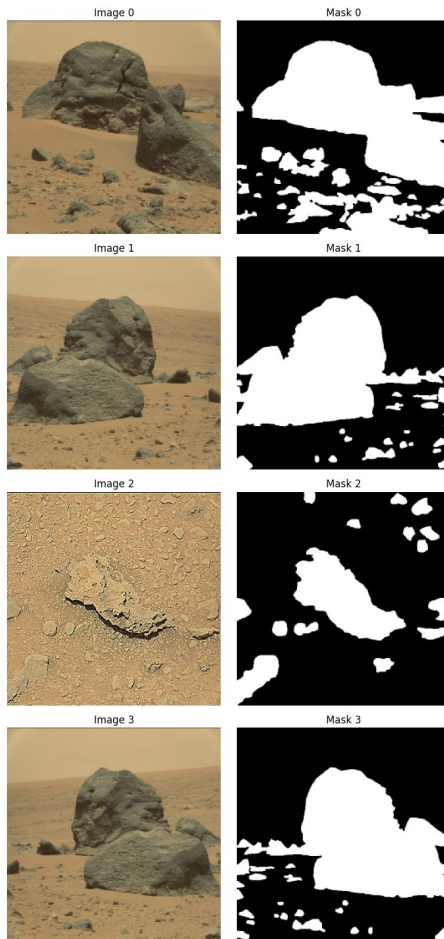
Associated Masks



Coded Masks To Single Channel

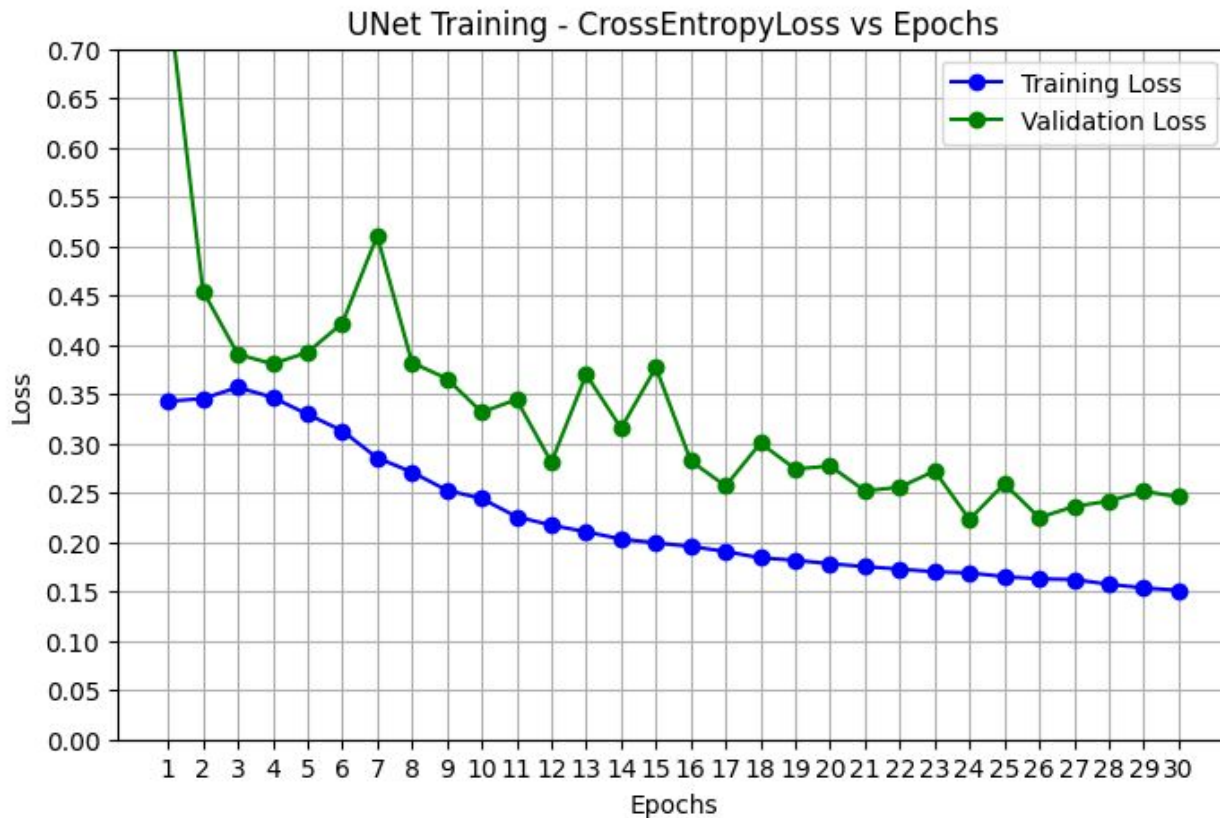
Target = 255

Background = 0



Training U-Net Image Segmentation Against Martian rocks

Training Loss & Validation after 30 epochs:



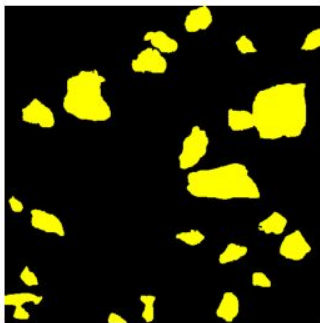
Mean Accuracy :
~0.89 for Test
subset size: 50
images

U-Net Inference on Martian Dataset

Original Image



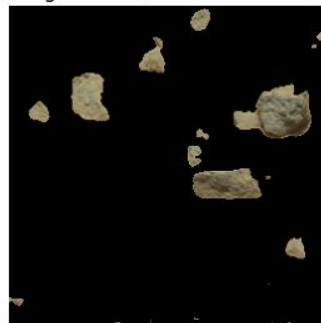
Ground Truth Mask



Predicted Mask



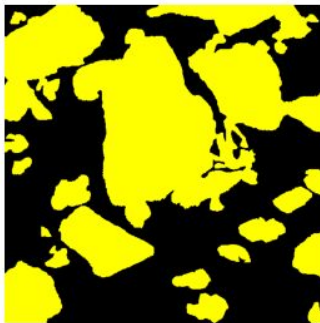
Segmented (Predicted Mask)



Original Image



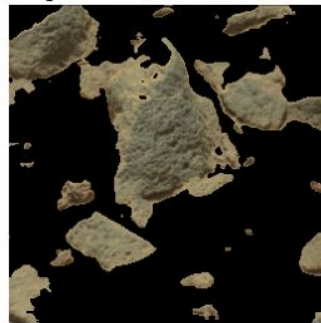
Ground Truth Mask



Predicted Mask



Segmented (Predicted Mask)



Conclusion & Further Work:

- In the future we need more real data. 30 images are not nearly enough.
- We also need carefully crafted synthetic data
- We need explore ensemble methods and any required post processing
- Other hazards should also be taken into account.